2 Why the design becomes (p_1, p_2, s)

Takeaway. Pages 1–2 isolate same-order strategic waiting in the exact one-driver benchmark. The market-design comparison now adds a fresh Poisson cohort in every second window and asks whether the platform should reprice the same catchment, widen the catchment, or do both.

2.1 Three strictly nested mechanism classes

Mechanism class	Commitment	Second-window core	What is added?
Flat fixed reach $\mathcal{M}_0$	$(p, p, 1)$	$N_2^C \sim \text{Pois}(m)$	No contingent price and no outer search.
Fixed-reach rescue $\mathcal{M}_F$	$(p_1, p_2, 1)$	$N_2^C \sim \text{Pois}(m)$	A committed rescue price in the same catchment.
Expanded-search rescue $\mathcal{M}_E$	(p_1, p_2, s), $s \geq 1$	$N_2^C \sim \text{Pois}(m)$	An independent outer annulus plus the same rescue price.

Thus "fixed reach" fixes the notification geography, not driver identities. Both rescue mechanisms contain time-homogeneous fresh arrivals in period 2. Their feasible sets are

$$\mathcal{M}_0 = \{(p, p, 1)\} \subset \mathcal{M}_F = \{(p_1, p_2, 1) : p_2 \geq p_1\} \subset \mathcal{M}_E = \{(p_1, p_2, s) : p_2 \geq p_1,\ 1 \leq s \leq \bar{s}\}. \qquad (5)$$

The flat benchmark has one optimized price; it is not a fixed-p_1 policy.

2.2 s is an area multiplier, not a participation share

Let spatially homogeneous fresh drivers form a Poisson point process. Normalize the core catchment to area one and let its radius be R_0. Committing to multiplier s means

$$R(s) = R_0\sqrt{s}, \qquad N_2^O(s) \sim \text{Pois}((s-1)m), \qquad N_2^C \perp N_2^O(s). \qquad (6)$$

Hence the expanded pool is $\text{Pois}(sm)$ and the core is its Poisson thinning with retention probability $1/s$. At $s = 4$, for example, area is quadrupled but radius only doubles.

2.3 The object plotted below

For market environment $\theta = (m, \beta, \delta)$, each line reports a separate equilibrium-constrained optimum

$$V_k(\theta) = \max_{\mu \in \mathcal{M}_k} \min_{a \in \mathcal{E}^{\text{WPBE}}(\mu;\theta)} M(\mu, a), \qquad k \in \{0, F, E\}. \qquad (7)$$

The nesting makes $V_E \geq V_F \geq V_0$; the substantive question is where each increment is large after all permitted controls are re-optimized.

3 Inner cutoff-WPBE, then outer mechanism design

Takeaway. The platform commits first. For every candidate mechanism, driver strategies, rider continuation, and Bayesian beliefs are solved as a complete cutoff-WPBE. Only then does the platform compare mechanisms.

3.1 Information and timing for the same focal order

Stage	Action	Information
0	Platform commits to $\mu = (p_1, p_2, s)$; a rider of value v posts iff $v \geq p_1$.	m, β, δ, the cost laws, and μ are public.
1	$N_1 \sim \mathrm{Pois}(m)$ incumbents see this order and choose accept p_1 or reject-and-wait.	Each sees the order, both prices, s, and own persistent cost c; counts and other costs are hidden.
2	After universal rejection, the rider chooses abandon, repeat at p_1, or rescue at p_2.	The rider sees failure but no driver count or future fresh responses; beliefs are induced by the equilibrium cutoff.
3	The chosen terminal notification is sent; fresh drivers see the order for the first time and may respond.	Rejectors and fresh volunteers enter one anonymous lottery. Pickup cost is paid only by the selected driver.

3.2 Winner-only pickup cost and terminal coverage

Write area rank as $u = (r/R_0)^2$. Core pickup is absorbed into $c \sim U[0, 1]$; the extra cost from expansion is

$$d(u) = \tau(\sqrt{u} - 1)_+.$$

A fresh driver's response payoff is $h(\lambda)[p - c - d(u)]$, where $h(z) = (1 - e^{-z})/z$. Because the same assignment probability multiplies payment and pickup cost,

$$\text{respond} \iff c + d(u) \leq p, \qquad e(p, s) = m \int_0^s F(p - d(u))\, du. \tag{8}$$

There is no sunk relocation cost and no fresh-entry fixed point. Given incumbent cutoff a, let $s_1 = 1, s_2 = s$ and

$$I_j(a) = m[F(p_j) - F(a)]_+, \quad \lambda_j(a) = I_j(a) + e(p_j, s_j), \quad C_j(a) = 1 - e^{-\lambda_j(a)}. \tag{9}$$

The rider compares 0, $C_1(\beta v - p_1)$, and $C_2(\beta v - p_2)$; let the resulting conditional action masses be $\eta_0(a), \eta_1(a), \eta_2(a)$.

3.3 Strict cutoff-WPBE test and the outer problem

For a proposed aggregate cutoff, a type-c incumbent's accept-minus-wait payoff is

$$D(c; a) = h(mF(a))(p_1 - c) - \delta e^{-mF(a)} \sum_{j=1}^{2} \eta_j(a) h(\lambda_j(a))(p_j - c)_+. \tag{10}$$

Together with the rider and fresh-driver strategies above, a is a cutoff-WPBE iff beliefs obey Bayes' rule and

$$D(c; a) \geq 0 \ \forall c < a, \qquad D(c; a) \leq 0 \ \forall c > a, \tag{11}$$

with indifference at an interior cutoff. Since D is piecewise affine with kinks only at p_1, p_2, the numerics verify (11) exactly at interval endpoints, not on a sampled type grid. The computation respects the nesting

$$\text{fix } p_1 \ \longrightarrow \ \max_{p_2, s} \ \min_{a \in \mathcal{E}^{\mathrm{WPBE}}} M \ \longrightarrow \ \max_{p_1}, \tag{12}$$

and enumerates the entire cutoff correspondence before applying the conservative equilibrium selection in (7).

4 Part 2: fixed-reach rescue

Takeaway. Fixed-reach rescue reprices the second window without changing its geography. It still has a new core Poisson cohort; its distinctive tool is the committed wedge $p_2 - p_1$, not the existence of fresh arrivals.

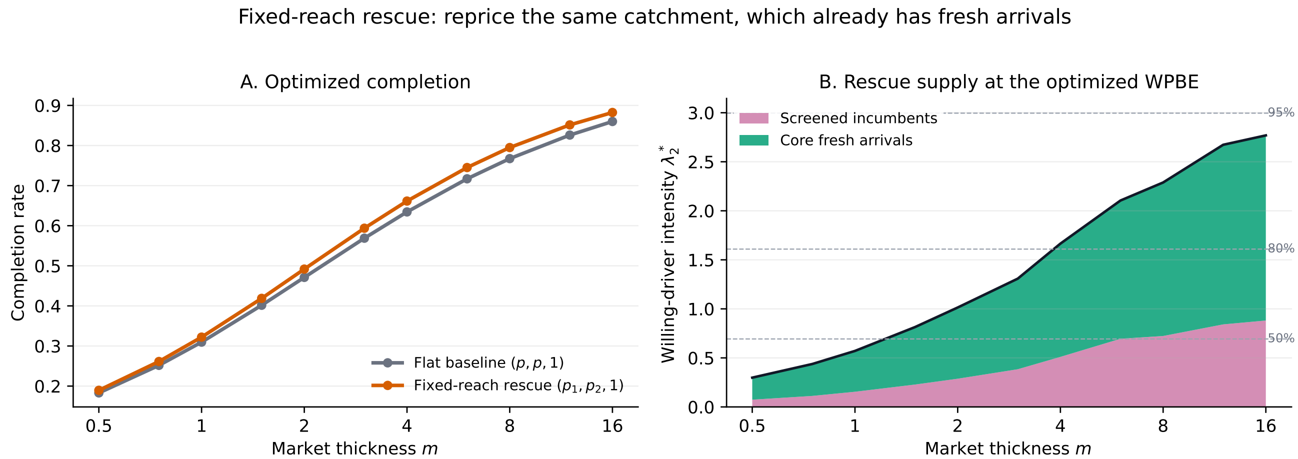

Figure 3: Fixed-reach rescue versus the flat two-window benchmark. Every point first solves the complete cutoff-WPBE and then re-optimizes the permitted mechanism. Right: second-window willing-driver intensity under the optimized fixed-rescue policy.

Under repeat, the terminal pool contains screened incumbents $m[F(p_1) - F(a)]_+$ and core fresh volunteers $e(p_1, 1)$. Under rescue, it contains $m[F(p_2) - F(a)]_+$ and $e(p_2, 1)$. For uniform costs,

$$e(p, 1) = mp. \tag{13}$$

Thus increasing p_2 works through two supply channels, but it also increases the continuation value of rejecting the focal order and may lower a.

At $(\beta, \delta, \tau) = (.8, .8, .25)$, the optimized fixed-rescue increment is only 0.65 percentage points at $m = .5$, peaks at 2.81 points near $m = 6$, and remains 2.24 points at $m = 16$. This hump is an equilibrium-design pattern: each orange point has its own (p_1^*, p_2^*) and its own WPBE cutoff.

Interpretation. In a very thin market, repricing a small catchment has little supply to work with. As m grows, the second-window core cohort and screened backup pool make contingent pay more useful. In a very thick market, the flat mechanism already completes most requests, limiting the remaining gain.

5 Part 3: expanded-search rescue

Takeaway. Expanded search preserves the core cohort and adds only the outer annulus. Distant drivers volunteer less often because the assigned winner, and only the winner, pays the incremental pickup cost.

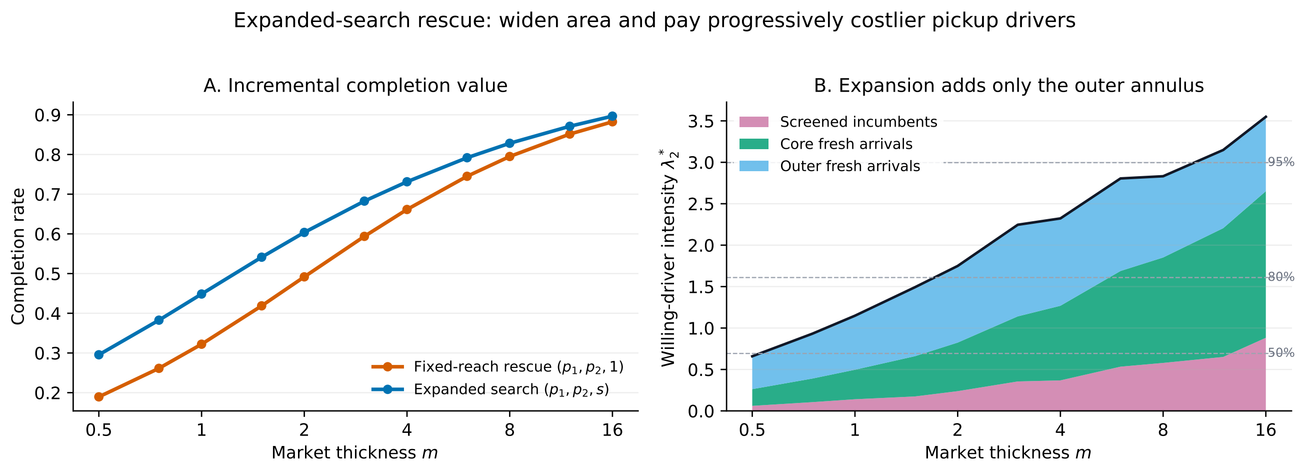

Figure 4: Incremental value and supply composition of expanded search. Each blue point re-optimizes (p_1, p_2, s) outside its complete cutoff-WPBE; the orange comparator separately re-optimizes $(p_1, p_2, 1)$.

For $F = U[0,1]$, $p \in [0,1]$, and $\tau > 0$, define

$$R(p,s) = \min\{\sqrt{s}, 1 + p/\tau\}.$$

Poisson thinning gives the closed-form fresh volunteer intensity

$$e(p,s) = mp + m(p+\tau)(R^2 - 1) - \frac{2m\tau}{3}(R^3 - 1). \tag{14}$$

Its marginal outer supply is

$$\frac{\partial e}{\partial s} = m[p - \tau(\sqrt{s} - 1)]_+, \qquad s^{\text{sat}}(p) = (1 + p/\tau)^2. \tag{15}$$

So s is a meaningful multiplier even without a platform search-cost term: beyond the saturation radius, extra notifications attract no willing drivers.

At $(\beta, \delta, \tau) = (.8, .8, .25)$, the incremental value of expanded search over optimized fixed rescue peaks at 12.64 percentage points near $m = 1$ and falls to 1.42 points at $m = 16$. The optimized multiplier declines from its cap $s^* = 4$ in thin markets to $s^* = 2.09$ at $m = 16$.

Interpretation. Search primarily repairs geographic scarcity. Price chooses which reached drivers are willing; s chooses who can first learn about the failed order. Both affect the incumbent's waiting option through terminal coverage and assignment competition, so the blue line is not a mechanical pool-size effect.

6 Under which market conditions does each tool work best?

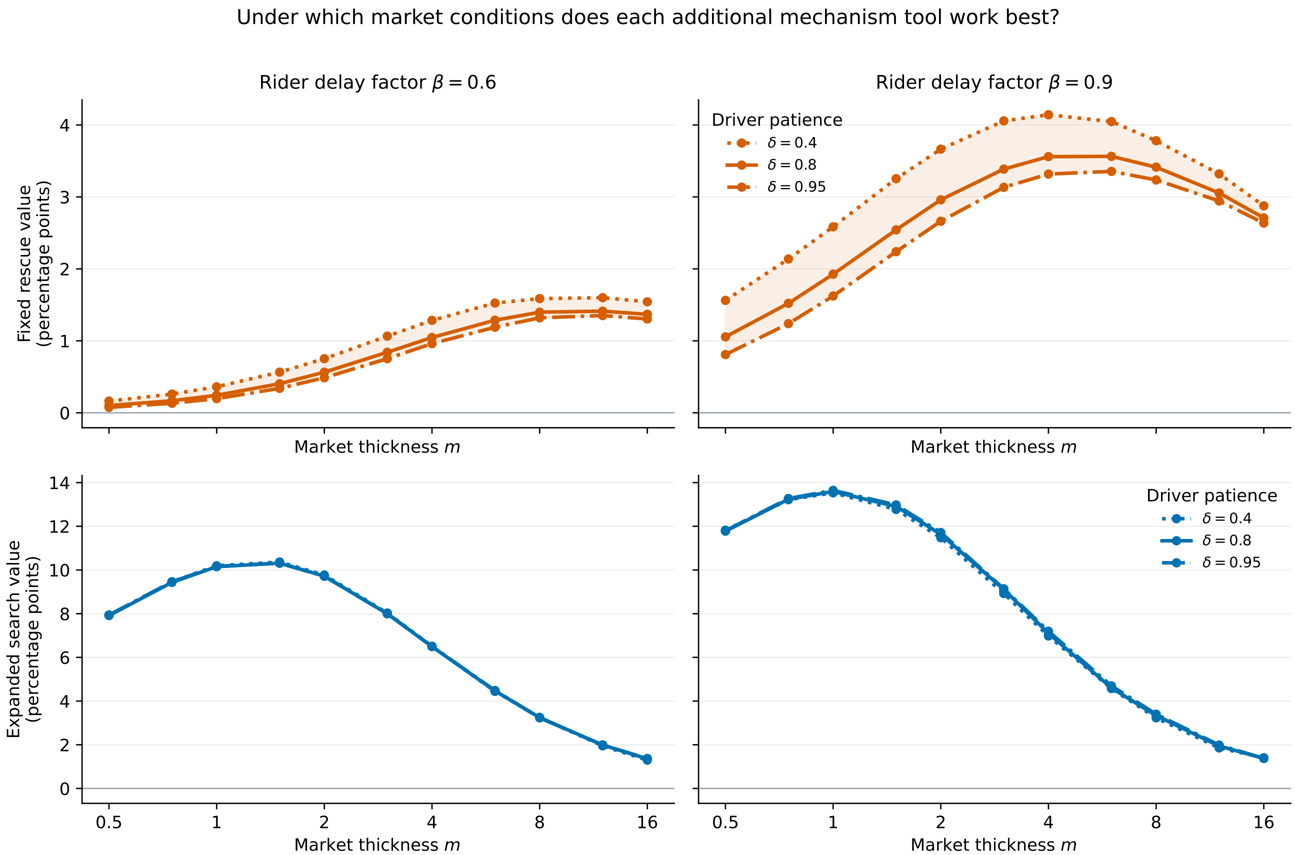

Figure 5: Top: value of adding fixed-reach rescue, $100(V_F - V_0)$. Bottom: incremental value of expanded search, $100(V_E - V_F)$. Every environment re-solves the inner WPBE and all outer policy variables; the shaded range is across the displayed δ values, not a confidence interval.

Condition	Fixed-reach rescue	Expanded-search increment
Thin m	A price wedge acts on a small core and is modest.	Largest value: the main failure is insufficient geographic reach.
Intermediate m	Value rises as the screened/core backup becomes usable.	Still material, but begins to decline after its thin-market peak.
Thick m	Hump-shaped: the flat benchmark eventually leaves little room.	Falls sharply because core coverage is already high.
Higher β	More riders continue after failure, making contingent pay more useful.	Also raises the level of search value.
Higher δ	More patient incumbents wait more; the price layer is noticeably affected.	Much less sensitive in this grid because added outer supply is fresh to the focal order.

Takeaway. The numerical policy reversal is not "pay in thick, search in thin" holding other controls fixed. It compares separately optimized, nested mechanism classes, each evaluated at its own cutoff-WPBE.

7 Closest papers and a short proposition chain

Takeaway. Search radius, two-round broadcasting, and spatial pickup frictions are known. The proposed contribution is their integration with a committed same-order price path, Bayesian continuation after universal rejection, and outer mechanism design over the full cutoff-WPBE correspondence.

Closest paper	Its mechanism / friction	Distinction here
Hu, Hu, Zhu (2022), M&SOM	Two-period surge and driver relocation; a driver pays a sunk cost before matching.	We use their fixed-plus-distance language, but pickup cost is assignment-contingent; the platform also commits to the notification reach of a failed focal order.
Yang, Qin, Ke, Ye (2020), TR-B	Joint matching interval and maximum pickup radius.	Supplies the radius trade-off, but has no committed (p_1, p_2) or strategic reject-and-wait response.
Qin, Yang, Liu (2025), TR-C	Two-round broadcast with round-specific radii under spatial Poisson arrivals.	Closest search design; it does not embed the radius choice in the same-order cutoff-WPBE and price mechanism.
Wang, Zhang, Zhang (2024), OR	Pickup begins after matching and occupies driver capacity; matching is controlled by pickup-time quality.	Supports winner-only pickup timing, but not order-level rescue pricing or Bayesian rider continuation.

7.1 Deliberately short proposition chain

1. **Spatial thinning.** Derive (14)–(15): $e_p > 0$, $e_s \geq 0$, and search saturates at $s^{\text{sat}}(p)$. This is the closed-form supply foundation, not the headline contribution.
2. **Cutoff-WPBE foundation.** Characterize $\mathcal{E}^{\text{WPBE}}(p_1, p_2, s)$ using (10)–(11), establish existence, and give conditions for uniqueness. The full correspondence, rather than one local root, is the constraint set faced by the platform.
3. **Market-condition allocation.** Give sufficient conditions under which the incremental optimum shifts from expanded reach in thin markets to contingent pay in thicker markets, accounting for rider continuation and incumbent holdout. The current figure is evidence, not yet a global theorem.

Numerical audit. The formal grid contains 77 environments and three mechanism classes (231 outer optima). Every optimum has one cutoff-WPBE and the same correspondence on grids 401, 801, and 1601. Independent dense re-optimization changes completion by at most 0.0112 percentage points. Thirteen model/solver tests pass; a 300,000-draw Poisson-thinning audit agrees with exact coverage within sampling error.

References. Hu, Hu, Zhu (2022), *M&SOM* 24(1), 91–109. Yang, Qin, Ke, Ye (2020), *TR-B* 131, 84–105. Qin, Yang, Liu (2025), *TR-C* 180, 105318. Wang, Zhang, Zhang (2024), *Operations Research* 72(3), 1278–1297.